LAPORAN WORKSHOP DEVELOPER OPERATIONAL
PENERAPAN DOCKER SWARM MINGGU KE-9

Oleh:

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BAB 1
HASIL PRAKTIKUM

1.1 Pengertian

Docker Swarm adalah fitur dari Docker yang digunakan untuk mengelola beberapa container dalam satu sistem cluster. Dengan Docker Swarm, aplikasi dapat dijalankan dalam beberapa instance (replica) untuk meningkatkan performa dan ketersediaan layanan.

1.2 Langkah-Langkah Praktikum

1. Update Package

```
rio@LAPTOP-K15MPFR2:/mnt/c/Users/RIO$ sudo apt update
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
43 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

2. Install Docker

```
rio@LAPTOP-K15MPFR2:/mnt/c/Users/RIO$ sudo apt install docker.io
[sudo] password for rio:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  bridge-utils containerd dns-root-data dnsmasq-base iptables libip4tc2 libip6tc2 libnetfilter-conntrack3
  libnfnetlink0 libnftables1 libnftnl11 nftables pigz runc ubuntu-fan
Suggested packages:
  ifupdown aufs-tools btrfs-progs cgroupfs-mount | cgroup-lite debootstrap docker-buildx docker-compose-v2 docker-doc
  rinse zfs-fuse | zfsutils firewalld
The following NEW packages will be installed:
  bridge-utils containerd dns-root-data dnsmasq-base docker.io iptables libip4tc2 libip6tc2 libnetfilter-conntrack3
  libnfnetlink0 libnftables1 libnftnl11 nftables pigz runc ubuntu-fan
0 upgraded, 16 newly installed, 0 to remove and 43 not upgraded.
Need to get 74.8 MB of archives.
After this operation, 283 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu noble/universe amd64 pigz amd64 2.8-1 [65.6 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble/main amd64 libip4tc2 amd64 1.8.10-3ubuntu2 [23.3 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble/main amd64 libip6tc2 amd64 1.8.10-3ubuntu2 [23.7 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble/main amd64 libnfnetlink0 amd64 1.0.2-2build1 [14.8 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble/main amd64 libnetfilter-conntrack3 amd64 1.0.9-6build1 [45.2 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble/main amd64 libnftnl11 amd64 1.2.6-2build1 [66.0 kB]
Get:7 http://archive.ubuntu.com/ubuntu noble/main amd64 iptables amd64 1.8.10-3ubuntu2 [381 kB]
Get:8 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libnftables1 amd64 1.0.9-1ubuntu0.1 [359 kB]
Get:9 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 nftables amd64 1.0.9-1ubuntu0.1 [69.8 kB]
Get:10 http://archive.ubuntu.com/ubuntu noble/main amd64 bridge-utils amd64 1.7.1-1ubuntu2 [33.9 kB]
Get:11 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 runc amd64 1.3.4-0ubuntu1~24.04.1 [9574 kB]
```

3. Restart Docker

```
rio@LAPTOP-K15MPFR2:/mnt/c/Users/RIO$ docker stop $(docker ps -q)
18f3be4584f2
7853202b656b
ccb6874b2ac7
499096cade38
9523ac3a070d
rio@LAPTOP-K15MPFR2:/mnt/c/Users/RIO$ docker rm $(docker ps-aq)
docker: unknown command: docker ps-aq

Run 'docker --help' for more information
docker: 'docker rm' requires at least 1 argument

Usage:  docker rm [OPTIONS] CONTAINER [CONTAINER...]

See 'docker rm --help' for more information
rio@LAPTOP-K15MPFR2:/mnt/c/Users/RIO$ docker rm $(docker ps -aq)
18f3be4584f2
7853202b656b
ccb6874b2ac7
499096cade38
9523ac3a070d
```

4. Inisialisasi Docker Swarm

```
rio@LAPTOP-K15MPFR2:/mnt/c/Users/RIO$ docker swarm init --advertise-addr 172.29.248.215
Swarm initialized: current node (xgfbeo7hsuz74fymo78olq97v) is now a manager.

To add a worker to this swarm, run the following command:

    docker swarm join --token SWMTKN-1-0d842gowb6dnpbxqm9d9z0q26l0lm760xscvb05c3bingbyjxj-4x0yb5taym7nzcizxwqylsp5d 172.
29.248.215:2377

To add a manager to this swarm, run 'docker swarm join-token manager' and follow the instructions.
```

5. Verifikasi Swarm

```
rio@LAPTOP-K15MPFR2:/mnt/c/Users/RIO$ docker info
Client:
 Version:    29.1.3
 Context:    default
 Debug Mode: false
 Plugins:
  ai: Docker AI Agent - Ask Gordon (Docker Inc.)
    Version:  v1.18.0
    Path:     /usr/local/lib/docker/cli-plugins/docker-ai
  buildx: Docker Buildx (Docker Inc.)
    Version:  v0.31.1-desktop.1
    Path:     /usr/local/lib/docker/cli-plugins/docker-buildx
  compose: Docker Compose (Docker Inc.)
    Version:  v5.0.2
    Path:     /usr/local/lib/docker/cli-plugins/docker-compose
  debug: Get a shell into any image or container (Docker Inc.)
    Version:  0.0.47
    Path:     /usr/local/lib/docker/cli-plugins/docker-debug
  desktop: Docker Desktop commands (Docker Inc.)
    Version:  v0.3.0
    Path:     /usr/local/lib/docker/cli-plugins/docker-desktop
  extension: Manages Docker extensions (Docker Inc.)
    Version:  v0.2.31
    Path:     /usr/local/lib/docker/cli-plugins/docker-extension
  init: Creates Docker-related starter files for your project (Docker Inc.)
    Version:  v1.4.0
    Path:     /usr/local/lib/docker/cli-plugins/docker-init
  mcp: Docker MCP Plugin (Docker Inc.)
    Version:  v0.39.1
```

Hasil:

Swarm aktif

Node sebagai Manager

6. Membuat Service

```
rio@LAPTOP-K15MPFR2:/mnt/c/Users/RIO$ docker service create --name diproperti-service -p 8085:80 nginx
xzx9mp72htr01vt9grafrkfou
overall progress: 1 out of 1 tasks
1/1: running   [==================================================>]
verify: Service xzx9mp72htr01vt9grafrkfou converged
```

7. Cek Service

```
rio@LAPTOP-K15MPFR2:/mnt/c/Users/RIO$ docker service ls
ID             NAME                MODE         REPLICAS   IMAGE          PORTS
xzx9mp72htr0   diproperti-service  replicated   1/1        nginx:latest   *:8085->80/tcp
```

8. Cek Container

```
rio@LAPTOP-K15MPFR2:/mnt/c/Users/RIO$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED             STATUS              PORTS     NAMES
9523ac3a070d   nginx:latest   "/docker-entrypoint.…"   About a minute ago  Up About a minute   80/tcp    diproperti-ser
vice.1.11jd288fx88anlqwdwvakjppd
```

9. Scaling Service

```
rio@LAPTOP-K15MPFR2:/mnt/c/Users/RIO$ docker service scale diproperti-service=5
diproperti-service scaled to 5
overall progress: 5 out of 5 tasks
1/5: running   [==================================================>]
2/5: running   [==================================================>]
3/5: running   [==================================================>]
4/5: running   [==================================================>]
5/5: running   [==================================================>]
verify: Service diproperti-service converged
```

10. Verifikasi Scaling

```
rio@LAPTOP-K15MPFR2:/mnt/c/Users/RIO$ docker service ps diproperti-service
ID             NAME                   IMAGE          NODE              DESIRED STATE   CURRENT STATE
R     PORTS
11jd288fx88a   diproperti-service.1   nginx:latest   LAPTOP-K15MPFR2   Running         Running 3 minutes ago

ksoa11mbz9tw   diproperti-service.2   nginx:latest   LAPTOP-K15MPFR2   Running         Running about a minute ago

p4byeijvvj8r   diproperti-service.3   nginx:latest   LAPTOP-K15MPFR2   Running         Running about a minute ago

s9qmf0giudxg   diproperti-service.4   nginx:latest   LAPTOP-K15MPFR2   Running         Running about a minute ago

j6z6130bs0dg   diproperti-service.5   nginx:latest   LAPTOP-K15MPFR2   Running         Running about a minute ago
```

Hasil:

Jumlah container bertambah menjadi 5

Semua container dalam status running

11. Pengujian

Akses melalui browser:

http://172.29.248.215:8085/85

Hasil:

Halaman Welcome to nginx berhasil ditampilkan

1.3 Hasil Praktikum

a. Docker Swarm berhasil dijalankan

b. Service berjalan dengan 3 replica awal

c. Scaling berhasil menjadi 5 replica

d. Semua container aktif dan berjalan normal

e. Aplikasi dapat diakses melalui browser